Problem solving with linear equations

ANSWER KEY

Core Concepts: Problem solving with linear equations

1 For the model $f(x) = 4x + 9$ in problem solving with linear equations, evaluate $f(2)$.

a. 17 ✓

b. 21

c. 8

d. 21

Substitute $x = 2$: $f(2) = 4(2) + 9 = 17$.

2 Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ **and** $x = 5$ ✓

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

Factoring $(x - 1)(x - 5) = 0$ yields the roots $x = 1$ and $x = 5$.

3 A data set recorded for problem solving with linear equations consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5 ✓

d. 26.5

The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

Procedural Fluency & Algebraic Manipulation

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4 For the model $f(x) = 4x + 9$ in problem solving with linear equations, evaluate $f(3)$.

a. 21 ✓

b. 25

c. 12

d. 25

Substitute $x = 3$: $f(3) = 4(3) + 9 = 21$.

5 Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

a. $x = 2$ and $x = 7$

b. $x = 2$ **and** $x = 6$ ✓

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Factoring $(x - 2)(x - 6) = 0$ yields the roots $x = 2$ and $x = 6$.

6 A data set recorded for problem solving with linear equations consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5 ✓

d. 26.5

The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

Graphical & Geometric Interpretation

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7 For the model $f(x) = 4x + 9$ in problem solving with linear equations, evaluate $f(4)$.

a. 25 ✓

b. 29

c. 16

d. 29

Substitute $x = 4$: $f(4) = 4(4) + 9 = 25$.

8 Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ **and** $x = 7$ ✓

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

Factoring $(x - 3)(x - 7) = 0$ yields the roots $x = 3$ and $x = 7$.

9 A data set recorded for problem solving with linear equations consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5 ✓

d. 26.5

The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

Exam-Style Problem Solving & Modeling

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10 For the model $f(x) = 4x + 9$ in problem solving with linear equations, evaluate $f(5)$.

a. 29 ✓

b. 33

c. 20

d. 33

Substitute $x = 5$: $f(5) = 4(5) + 9 = 29$.

11 Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ **and** $x = 8$ ✓

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

Factoring $(x - 4)(x - 8) = 0$ yields the roots $x = 4$ and $x = 8$.

12 A data set recorded for problem solving with linear equations consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5 ✓

d. 26.5

The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

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